RW–10676

Sub. Code
511701

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Maths

OBJECT ORIENTED PROGRAMMING AND C++

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the use of Encapsulation?

2. What are Constructors?

3. List the features of member functions.

4. What is the difference between object and variable?

5. List the uses of pointers?

6. What is the use of 'new' operator?

7. What is the significance of Polymorphism?

8. What is the difference between operator overloading and function overloading?

9. Define Pure Virtual function.

10. What is dynamic binding?

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) List the features of object oriented programming.

Or

(b) List the application areas of OOP.

12. (a) What is inline Function? Explain briefly.

Or

(b) Explain the constructor with suitable example.

13. (a) Briefly explain the dynamic constructor with example.

Or

(b) Explain any five functions supported by string class with examples.

14. (a) List the rules for overloading operators.

Or

(b) With suitable example, explain Operator overloading.

15. (a) Explain Virtual functions with example.

Or

(b) Write a short note on Abstract class.

Part C (3 × 10 = 30)

Answer any **three** questions only.

16. Write a C++ program to read three numbers from the keyboard and display the Biggest value on the screen.

17. Write a C++ program to do the following :

(a) Area of circle

(b) Circumference of the circle

(c) Volume of the cylinder

(d) Surface area of the closed cylinder

(e) Volume of sphere.

18. What is Pointer? Write a C++ program to display the address of the variable.

19. With suitable example C++ program explain functin overloading.

20. Explain about pure virtual functions and virtual Base Class with examples.

RW–10455

Sub. Code
511102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

REAL ANALYSIS

(CBCS – 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give an example to show that arbitrary union of closed sets need not be closed.

2. Show that $\left\{\frac{1}{n}\right\} \cup \{0\}$ is compact.

3. If f, g are continuous real-valued functions on $\mathbf{R}$ and if $f = g$ on Q, the set of rationals, then show that $f \equiv g$ on $\mathbf{R}$.

4. If $f : [a,b] \to \mathbf{R}$ is one-one continuous then show that there exists $c, d \in \mathbf{R}$ such that $f([a,b]) = [c,d]$.

5. If $f(x)$ is defined for all real x and if for all real $x, y, |f(x) - f(y)| \le (x-y)^2$, then show that f is a constant.

6. Show that the function $f : [0, 2\pi] \to \mathbf{R}^2$ defined by $f(t) = (\cos t, \sin t)$ does the satisfy Mean-Value theorem.

7. State the necessary and sufficient condition for a bounded function to be Riemann stieltjes integrable.

8. Prove that $\int_a^b d\alpha = \alpha(b) - \alpha(a)$.

9. Show that $\sum \frac{x}{1+n^2 x}$ converges uniformly on R.

10. If $f_n(x) = x^n, 0 \le x \le 1$, then show that f_n does not converge uniformly on $[0,1]$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that compact subset of a metric space set is closed.

Or

(b) Show that every bounded infinite subset of $\mathbf{R}^k$ has a limit point in $\mathbf{R}^k$.

12. (a) Show that continuous image of a connected set is connected.

Or

(b) If f is a continuous mapping of a metric space X into a metric space Y then prove that $f(\overline{E}) \subset \overline{f(E)}$.

13. (a) If f is a real differentiable function on $[a,b]$ and if $f'(a) < \lambda < f'(b)$, then show that there exists $x \in (a,b)$ such that $f'(x) = \lambda$.

Or

(b) State and prove Chain rule.

14. (a) If α increases monotonically on $[a, b]$ and if P^* is the refinement of the partition P of $[a, b]$ then show that $L(P,f,\alpha) \leq L(P^*,f,\alpha)$.

Or

(b) If $f \in R$ on $[a,b], x \in [a,b]$ then show that the Function F defined by $F(x) = \int_a^x f(x)dx$ is differentiable whenever f is differentiable.

15. (a) If $f_n \to f$ uniformly and if each f_n is continuous at a then show that f is continuous at a.

Or

(b) State and prove Weierstrass M-test.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that a subset E of **R** is connected iff for all $x, y \in E$ and $x < z < y$, then $z \in E$.

17. If f is a continuous mapping of a compact metric space X into a metric space Y, then show that f is uniformly continuous.

18. State and prove L' Hospital's rule.

19. If f is a bounded real function on $[a, b]$, α increases monotonically and if $\alpha' \in R$ on $[a, b]$ then show that $f \in R(\alpha)$ iff $f\alpha' \in R$ and hence show that $\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x)dx$.

20. If each f_n, is real-valued differential function in (a,b), if there is at least one $x_0 \in (a,b)$ the sequence $\{f_n(x_0)\}$ converges and if $\{f_n'\}$ converges uniformly on (a,b) then show that $\{f_n\}$ converges uniformly to a function f and $f'(x) = lin_{n\to\infty} f_n'(x)$.

RW–10456

Sub. Code
511103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

ORDINARY DIFFERENTIAL EQUATIONS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each question carries 2 marks.

1. Define linearly independent function.

2. Let $\phi_1(x) = x^2, \phi_2(x) = x|x|$. Find the Wronskian.

3. One solution of $x^3y''' - 3x^2y'' + 6xy' - 6y = 0$ for $x > 0$ is $\phi_1(x) = x$. Find a basis for the solution for $x > 0$.

4. Define Chebyshev equations.

5. Find all solutions of the equation $x^2y'' + 2xy' - 6y = 0 : x > 0$.

6. Obtain two linearly independent solutions of the following equation which are valid near $x = 0 : x^2y'' + 2x^2y' - 2y = 0$.

7. Explain the method of successive approximation.

8. Define exact equation with example.

9. What do you mean by non-local existence of solutions?

10. Consider the problem $y' = y + \lambda x^2 \sin y, y(0) = 1$ where λ is some real parameter, $|\lambda| \le 1$. Show that a solution ψ of this problem exists for $|x| \le 1$.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Prove that two solutions ϕ_1, ϕ_2 of $L(y) = 0$ are linearly independent on an interval I if and only if $W(\phi_1, \phi_2)(x) \neq 0$ for all x in I.

Or

(b) Let b be continuous on an interval I. Every solution ψ of $L(y) = b(x)$ on I can be written as $\psi = \psi_p + c_1 \phi_1 + c_2 \phi_2$, where ψ_p is a particular solution, ϕ_1, ϕ_2 are two linearly independent solutions of $L(y) = 0$ and c_1, c_2 are constants. Prove that a particular solution ψ_p is given by

$$\psi_p = \int_{x_0}^{x} \frac{[\phi_1(t)\,\phi_2(x) - \phi_1(x)\,\phi_2(t)]\,b(t)\,dt}{W(\phi_1, \phi_2)(t)}.$$

Also conversely prove that every such ψ is a solution of $L(y) = b(x)$.

12. (a) Let ϕ_1 be a solution of $L(y) = 0$ on an interval I, and suppose $\phi_1(x) \neq 0$ on I. If $v_2, \ldots. v_n$ is any basis on I for the solutions of the linear equation

$$\phi_1 v^{(n-1)} + \ldots.. + \left[n\phi_1^{(n-1)} + (n-1)\,\phi_1^{(n-2)} + \ldots. + a_{n-1}\phi_1\right]$$

$v = 0$ of order $n-1$, and if $v_k = u_k', (k = 2, \ldots., n)$, then prove that $\phi_1, u_2\,\phi_1, \ldots., u_n\,\phi_1$ is a basis for the solutions of $L(y) = 0$ on I.

Or

(b) Show that the coefficient of x^n in $P_n(x)$ is $\dfrac{(2n)!}{2^n (n!)^2}$.

13. (a) Find the singular points of the following equation, and determine those which are regular singular points : $x^2y'' + \left(x + x^2\right)y' - y = 0$.

Or

(b) Prove that the series J_0 and K_0 converge for $|x| < \infty$.

14. (a) Find all real-valued solutions of the equation $yy' = x$.

Or

(b) Consider the initial value problem $y' = 3y + 1$, $y(0) = 2$. Show that all the successive approximations $\phi_0, \phi_1, \ldots.$ exists for all real x.

15. (a) Let f, g be continuous on R, and suppose f satisfies a Lipschitz condition there with Lipschitz constant K. Let ϕ, ψ be solutions of $y' = f(x,y), y(x_0) = y_1$ and $y' = g(x,y), y(x_0) = y_2$ respectively on an interval I containing x_0, with graphs contained in R. If the inequalities $|f(x,y) - g(x,y)| \le \in$, $((x,y) \text{ in } R)$, and $|y_1 - y_2| \le \delta$ are valid , then prove that

$$|\phi(x) - \psi(x)| \le \delta\, e^{K|x-x_0|} + \frac{\in}{K}\left(e^{K|x-x_0|} - 1\right) \text{ for all } x \text{ in } I.$$

Or

(b) Let f be a real-valued continuous function on the strip $S : |x| \le a, |y| < \infty$, $(a > 0)$ and suppose f satisfies a Lipschitz condition on S. Show that the solution of the initial value problem $y'' + \lambda^2 y = f(x,y), y(0) = 0$, $y'(0) = 1, (\lambda > 0)$ is unique.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. Let $r_1.....,r_s$ be the distinct roots of the characteristic polynomial p, and suppose r_i has multiplicity m_i (thus $m_1 + m_2 + ... + m_s = n)$). Prove that the n solutions $e^{r_1 x}, xe^{r_1 x}, ...x^{m_1 - 1}e^{r_1 x}; e^{r_2 x}, xe^{r_2 x}, ...x^{m_2 - 1}e^{r_2 x}, ...;$ $e^{r_s x}, xe^{r_s x}, ...x^{m_s - 1}e^{r_s x}$ of $L(y) = 0$ are linearly independent on any interval *I*.

17. State and prove existence theorem for linear equation with analytic coefficients.

18. Find all solutions ϕ of the form $\phi(x) = |x|^r \sum_{k=0}^{\infty} C_k x^k$, $(|x| > 0)$, for the following equation $3x^2 y'' + 5xy' + 3xy = 0$.

19. Let M, N be two real-valued functions which have continuous first partial derivatives on some rectangle $R: |x - x_0| \le a, |y - y_0| \le b$. Then prove that the equation $M(x,y) + N(x,y)y' = 0$ is exact in R if and only if $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$ in R.

20. Let f be a continuous vector-valued function defined on $S: |x - x_0| \le a, |y| < x$, $(a > 0)$, and satisfy there a Lipschitz function. Then prove that the successive approximations $\{\phi_k\}$ for the problem $y' = f(x,y), y(x_0) = y_0, (|y_0| < \infty)$, exist on $|x - x_0| \le a$, and converge there to a solution ϕ of this problem.

RW–10457

Sub. Code
511104

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

ANALYTIC NUMBER THEORY

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. If $a \mid bc$ and if $(a, b) = 1$ then prove that $a \mid c$.

2. Define Mobius function.

3. State Euler's summation formula.

4. Find $[2x] - 2[x]$.

5. Define Chebyshev's θ — function.

6. State Selberg asymptotic formula.

7. Assume $a \equiv b \pmod{m}$. If $d \mid m$ and $d \mid a$ then prove that $d \mid b$.

8. Define reduced residues system modulo m.

9. Define Legendre's symbol.

10. Find $(219 \mid 383)$.

Part B (5 × 5 = 25)

Answer **all** questions, by choosing (a) or (b).

Each question carries 5 marks.

11. (a) State and prove fundamental theorem of arithmetic.

Or

(b) If $n \geq 1$ then prove that $\phi(n) = \sum_{d|n} \mu(d)\frac{n}{d}$.

12. (a) If $x \geq 1$ and $\alpha > 0$, $\alpha \neq 1$, then prove that $\sum_{n \leq x} \sigma_\alpha(n) = \frac{\zeta(\alpha+1)}{\alpha+1} x^{\alpha+1} + O(x^\beta)$ where

$\beta = \max\{1, \alpha\}$.

Or

(b) For all $x \geq 1$ prove that $\left|\sum_{n \leq x} \frac{\mu(n)}{n}\right| \leq 1$ with equality holding only if $x < 2$.

13. (a) State Able's identity.

Or

(b) Prove that there is a constant A such that $\sum_{p \leq x} \frac{1}{p} = \log\log x + A + O\left(\frac{1}{\log x}\right)$ for all $x \geq 2$.

14. (a) If $(a,b) = d$ then prove that there exist integers x and y such that $ax + by = d$.

Or

(b) State and prove Wilsons theorem.

15. (a) State and prove Gauss lemma.

Or

(b) If P and Q are positive odd integers with $(P,Q)=1$ then prove that $(P \mid Q)(Q \mid P)=(-1)^{(P-1)(Q-1)/4}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. For all $n \geq 1$ than prove that $\phi(n)=n\prod_{p|n}\left(1-\frac{1}{p}\right)$.

17. Prove that the set of all lattice points visible from the origin has density $\frac{6}{\prod^2}$.

18. Prove that $\lim_{x\to\infty}\left(\frac{M(x)}{x}-\frac{H(x)}{x\log x}\right)=0$.

19. For any prime $p \geq 5$, prove that $\sum_{k=1}^{p-1}\frac{(p-1)!}{k} \equiv 0(\bmod p^2)$.

20. If p is an odd positive integer then prove that $(-1 \mid P)=(-1)^{(P-1)/2}$ and $(2 \mid P)=(-1)^{(P^2-1)/8}$.

RW–10458

Sub. Code
511513

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

OBJECT ORIENTED PROGRAMMING AND C++

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Give any four applications of OOPS.

2. What do you mean by dynamic initialization of variables?

3. Define : Data Encapsulation.

4. What is the benefit of Inheritance?

5. What is the advantages of pointers?

6. What is the function of delete( ) operator?

7. What is Function overloading?.

8. What is Operator overloading?

9. Define Abstract Class.

10. What is the use of dynamic binding?

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Briefly explain the manipulators.

Or

(b) Describe bit-wise operators available in C++.

12. (a) With suitable example explain the constructor.

Or

(b) Explain the access specifier in detail.

13. (a) What is this pointer? Explain its use.

Or

(b) Explain copy constructor with suitable C + + coding.

14. (a) Explain the different forms of Polymorphism.

Or

(b) List the pitfalls of operator overloading.

15. (a) Explain pure virtual function with an example.

Or

(b) Write a short note on run-time polymorphism.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the C + + data types in detail.

17. Explain the declaration of class in C + +. How will you define the member function of a class? Explain.

18. Discuss the string functions in detail.

19. Discuss about the Over loadable Operators.

20. Write a C ++ probram with a mother class forest-animal inside it define a name, age and height variables, and set value() function. Then create two base variables Deer and Elephant which write a message telling the age, the name, height and giving some extra information (e.g. weight.

RW–10459

Sub. Code
511301

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

MECHANICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define degrees of freedom.

2. Define virtual displacement.

3. Write the standard form of Lagrange's equation for a nonholonomic system.

4. Write the Lagrangian function.

5. Write the second variation of f at the stationary point q_o.

6. Define the Hamiltonian function.

7. Define the integrals of the motion.

8. State the Jacobi's theorem.

9. Define Mathieu transformation.

10. Write the bilinear covariant.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain configuration space.

Or

(b) Derive Lagrangian form of D'Alermbert's principle.

12. (a) Illustrate the Routhian method for the Kepler problem.

Or

(b) Explain: Liouville's system.

13. (a) Discuss: The brachistochrome problem.

Or

(b) Obtain the orbit for the Kepler problems by using the Jacobi form of the principle of least action.

14. (a) Explain : Pfaffian Differential forms.

Or

(b) Analyze the Kepler problem by using the Hamilton-Jacobi method.

15. (a) Show that the following transformation is canonical:

$$\theta = \tfrac{1}{2}\left(q^2 + p^2\right), \quad p = -\tan^{-1} q/p$$

Or

(b) Discuss : momentum transformation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) A particle of mass m is suspended by a massless wire of length $r = a + b\cos wt$ $(a > b > 0)$ to form a spherical pendulum. Find the equations of motion.

    (b) Find the expressions for the kinetic energy and the generalized momenta.

17. Consider the spherical pendulum. Reduce the problem to quadratures and obtain the integrals of the motion.

18. Derive the Jacobi's form of the principle of least action.

19. State and prove Jacobi's theorem.

20. Discuss the generating function for the transformation.

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RW–10460

Sub. Code
511302

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

TOPOLOGY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Show that intersection of two topologies on a set X is a topology.

2. Define order topology on a set X with a simple order.

3. Find the inverse of the function $f : R \to R$ given by $f(x) = 3x + 1$.

4. Give an example to show that a subspace of a quotient space need not be a quotient space.

5. Write the necessary and sufficient conditions for a subset of R to be a connected set.

6. Show that R with cofinite topology is compact.

7. Show by an example that subspace of a Lindelof space is not Lindelof.

8. Show that second countability implies separability.

9. Define completely regular space.

10. State Tychonoff theorem.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) If A is a subset of topological space X and if for each $x \in A$ there is an open set U containing x such that $U \subset A$, then show that A is open in X.

Or

(b) If A and B are subsets of a topological space X then prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

12. (a) If X is a metric space with metric d then show that $\overline{d}$ defined by $\overline{d}(x, y) = \min\{d(x, y), 1\}$ induces the same topology as d.

Or

(b) Let X and Y be connected spaces. Then show that $X \times Y$ is connected.

13. (a) Prove that every compact subspace of a Hausdorff space is closed.

Or

(b) Prove that a metric space is compact iff it is complete and totally bounded.

14. (a) Prove that every metrizable space is normal.

Or

(b) Prove that every compact Hausdorff space is normal.

15. (a) Let X be a Hausdorff space. Then show that X is locally compact iff given $x \in X$ and given neighborhood of x, there is a neighborhood V of x such that $\overline{V}$ is compact and $\overline{V} \subset U$.

Or

(b) Prove that a product of completely regular spaces is completely regular.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) If $\mathcal{C}$ is a collection of open sets in a topological space X such that for each open set U of X and each x in U there is an element C of $\mathcal{C}$ such that $x \in C \subset U$, then prove that $\mathcal{C}$ is a basis for the topology of X.

(b) Show that the topologies of R_l and R_k are strictly finer than standard topology on R.

17. Show that the topologies on R^n induced by the Euclidean metric d and the square metric ρ are the same as the product topology on R^n.

18. Prove that the product of finitely many compact spaces is compact.

19. State and prove Urysohn's lemma.

20. Let X be a completely regular space. Then show that there exists a compactification Y of X such that every bounded continuous map $f : X \to R$ extends uniquely to a continuous map of Y into R.

RW–10461

Sub. Code
511303

M.Sc. (Mathematics) DEGREE EXAMINATION, NOVEMBER 2015

DIFFERENTIAL GEOMETRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each question carries 2 marks.

1. Prove that a necessary and sufficient condition that a curve to be a straight line is that $K = 0$ at all points.

2. What is the curvature and torsion of the curve $x^2 + y^2 = 25$?

3. Define involutes and evolutes.

4. For the osculating sphere, write the radius of spherical curvature and the position vector of centre of curvature.

5. Define the axis and the pitch of a helicoid.

6. Find the orthogonal trajectories of tire circle $r = a\cos\theta$.

7. Show that a curve on a plane is a geodesic if and only if, it is a straight line.

8. Define geodesic curvature and prove that the geodesic curvature of a geodesic is zero.

9. State Rodrigue's formula.

10. Define polar developable and rectifying developable of a space curve.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

Each question carries 5 marks.

11. (a) Obtain a formula for the arc length of a path given in terms of a parameter and also show that the length of one complete turn of the helix. $r=(a\cos u, b\sin u, bu)\, a>0, -\infty<u<\infty$ is $2\pi C$, where $C=\sqrt{a^2+b^2}$.

Or

(b) Prove that the length of the common perpendicular distance d between the tangents at two neighboring points with the arcual distance s between them is approximately $d=\frac{kTs^3}{12}$.

12. (a) If the radius of spherical curvature is constant, prove that the Curve either lies on a sphere (or) has constant curvature.

Or

(b) Find the intrinsic equation of the curve, $\bar{\gamma}=(ae^u\cos u, ae^u\sin u, be^u)$.

13. (a) On the paraboloid $x^2 - y^2 = z$, find the orthogonal trajectories of the sections by the planes z = constant.

Or

(b) Find the surface of revolution which is isometric with the region of the right helicoid.

14. (a) Prove that the curves of the family $\frac{v^3}{u^2}$ = constant are geodesics on a surface with metric $v^2 du^2 - 2uv\,du\,dv + 2u^2 dv^2, (u > 0, v > 0)$.

Or

(b) Obtain the geodesic equations using Christoffel symbols of first kind.

15. (a) Derive the equation giving the principal curvatures.

Or

(b) If there is a surface of minimum area passing through a closed space curve, prove that it is necessary a surface of zero mean curvature.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. Obtain the curvature and torsion of the curve of intersection of the two quadratic surfaces $ax^2 + by^2 + cz^2 = 1, a'x^2 + b'y^2 + c'z^2 = 1$.

17. Find the curvature and torsion of the involutes. Also show that the torsion of the involute is equal to $\frac{\rho(\sigma\rho^1 - \sigma^1\rho)}{(\rho^2 + \sigma^2)(c - \rho)}$.

18. Prove that two surfaces of the same constant curvature are locally isometric.

19. State and prove Gauss-Bonnet theorem.

20. Prove that a necessary and sufficient condition that a curve on a surface be a line of curvature is that the surface normals along the curve form a developable.

RW–10462

Sub. Code
511505

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

GRAPH THEORY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each question carries 2 marks.

1. Prove that in any graph the number of odd degree vertices is even.

2. What is the number of edges in K_6? Justify.

3. Define Euler graph and Hamiltonian graph. Give an example of an Euler graph which is not Hamiltonian.

4. Define the closure of a graph. What is the closure of the graph K_n?

5. Write down the relationship between maximum matching and perfect matching.

6. Define Ramsay number. What is $r(3, 3)$?

7. Define edge chromatic: number and give an example.

8. State Hajo's theorem.

9. State Euler's formula and verify it for $K_{2,3}$.

10. Define a tournament and give an example.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

Each question carries 5 marks.

11. (a) If G and H are isomorphic, then show that $v(G) = v(H)$ and $\varepsilon(G) = \varepsilon(H)$. Give an example to show that the converse is false.

Or

(b) Prove that every nontrivial loopless connected graph has at least two vertices which are not cut vertices.

12. (a) With usual notations prove that $k \leq k' \leq \delta$.

Or

(b) Prove that a nonempty connected graph is Eulerian if it has no vertex of odd degree.

13. (a) Define perfect matching and prove that K_n has a perfect matching if and only if n is even.

Or

(b) Prove that every 3-regular graph without cut edges has a perfect matching.

14. (a) Show that the Peterson graph is 4-edge chromatic.

Or

(b) If G is a connected simple graph and is neither an odd cycle nor a complete graph, then prove that $\chi \leq \Delta$.

15. (a) If G is a connected plane graph show that $v - \varepsilon + \phi = 2$.

Or

(b) Prove that a digraph contains a directed path of length $\chi - 1$.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

Each question carries 10 marks.

16. State and prove Cayley's formula.

17. Define the graph $H_{m,n}$ n and prove that it is m-connected.

18. Prove Erdös theorem on Ramsey number.

19. State and prove Turan's theorem.

20. State and prove the five color theorem.

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RW–10463

Sub. Code
511703

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

IMAGE PROCESSING AND PATTERN RECOGNITION

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Define Sampling.

2. What is multi-image operation?

3. What is meant by edge detection?

4. What is the use of raster screen?

5. Define quadrees.

6. What is Fractal Compression?

7. Define pattern recognition.

8. Give any three examples for automatic pattern recognition systems.

9. What is Single Prototype?

10. Define Cluster.

Section B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write a short note on gray level transformation

Or

(b) Discuss the image processor.

12. (a) Explain opening and closing operations.

Or

(b) Explain about pyramid edge detection.

13. (a) Give brief explanation of spatial compression.

Or

(b) Explain about Contour coding.

14. (a) Explain the System design.

Or

(b) Briefly explain Cluster seeking.

15. (a) How to measure the similarity clustering criteria?

Or

(b) Explain the minimum distance pattern classification.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain image model with diagrams.

17. Discuss (a) Basic edge detection (b) Crack edge detection.

18. Discuss the Real Time image transmission.

19. Describe the decision functions in detail.

20. Explain the pattern classification by distance functions.

RW–10625

Sub. Code
511201

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

ALGEBRA — II

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. In a vector space show that $\alpha(v-w)=\alpha v-\alpha w$.

2. Define the following terms :

   (a) Inner product space.

   (b) Otrhonormal set.

3. Show that the mapping $\psi: F[x]\to F(a)$ defined by $h(x)\psi=h(a)$ is a homomorphism.

4. If F is of characteristic 0 and $f(x)\in F[x]$ is such that $f'(x)=0$, then prove that $f(x)=\alpha\in F$.

5. Let k be the field of complex numbers and let F be the field of real numbers. Find $G(K,F)$.

6. Define the following terms :

   (a) Galois group.

   (b) Solvable group.

7. Let $S,T \in A(v)$ and $m(s)=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $m(T)=\begin{pmatrix} -1 & 0 \\ 2 & 3 \end{pmatrix}$.

Find $m(ST)$.

8. If $T \in A(v)$ is nilpotent, then prove that $\alpha_0 + \alpha_1 T + ... + \alpha_m T^m$, where the $\alpha_i \in F$.

9. Check whether the matrix is unitary or not $\begin{pmatrix} 1 & 2-i \\ 2-i & i \end{pmatrix}$.

10. If the finite field F has p^m elements then prove that every $a \in F$ satisfies $a^{pm} = a$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) If V is the internal direct sum of $u_1, u_2, ... u_n$, then prove that V is isomorphic to the external direct sum of $u_1, u_2, ... u_n$.

Or

(b) Write the usual notations, prove that $A(A(W)) = W$.

12. (a) If L is an algebraic extension of k and if k is an algebraic extension of F, then prove that L is an algebraic extension of F.

Or

(b) If α is constructible then prove that α lies in some extension of the rationals of degree a power of 2.

13. (a) Let $G = S_n$, where $n \geq 5$; then prove that $G^{(k)}$ for $k = 1, 2...., $ contains every $3-$cycle of s_n.

Or

(b) What are the Galois group over Q of the polynomial $p(x) = x^5 + 5x^4 + 10x^3 + 10x^2 - x - 2$.

14. (a) Given the matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix} \in F_3$. Prove that $A^3 - 6A^2 + 11A - 6I = 0$.

Or

(b) Prove that two nilpotent linear transformations are similar if and only if they have the some invariants.

15. (a) If $T \in A(v)$ is such that $(vT, v) = 0$ for all $v \in V$, then prove that $T = 0$.

Or

(b) Show that the multiplicative group of nonzero elements of a finite field is cyclic.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let V be a finite – dimensional inner product space. Prove that V has an Orthonormal set as a basis.

17. If L is a finite extension of k and if k is a finite extension of F, then prove that L is a finite extension of F. Also deduce that $[L:F] = [L:K][K:F]$.

18. Let k be the splitting field of $f(x)$ in $f[x]$ and let $p(x)$ be an irreducible factor of $f(x)$ in $f[x]$. If the roots of $p(x)$ are $\alpha_1, \alpha_2 .....\alpha_r$, then prove that for each i there exists an automorphism σ_i in $G(K,F)$ such that $\sigma_i(\alpha_1) = \alpha_i$.

19. If the matrix $A \in F_n$ has all its characteristic roots in F, then prove that there is a matrix $C \in F_n$ such that CAc^{-1} is a triangular matrix.

20. (a) For all $A, B \in F_n$. prove that following.

    (i) $(A')' = A$.

    (ii) $(A+B)' = A' + B'$.

    (iii) $(AB)' = B'A'$.

    (b) Show that $T \in A(v)$ is unitary if and only if $TT^* = 1$.

RW–10626

Sub. Code
511202

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

MEASURE AND INTEGRATION

(2011 Onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define the concept of outer measure $m^*(A)$.

2. Show that every σ–algebra is a σ–ring.

3. Prove that continuous functions are measurable.

4. Give an example of a function such that $|f|$ is measurable but f is not.

5. Show that $f = f^+ - f^-$.

6. What is meant by Riemann integrable over $[a, b]$?

7. Define a signed measure with an example.

8. Define the following terms :

   (a) Mutually singular

   (b) Absolutely continuous.

9. If $\{A_i\}$ is a monotone sequence of sets, then prove that $\lim A_i^y = (\lim A_i)^y$.

10. Let μ and γ be complete measures. Show that $\mu \times \gamma$ need not be complete.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Show that if $F \in \mathfrak{M}$ $(F \Delta G) = 0$, then G is measurable.

Or

(b) If μ is a σ–finite measure on R then prove that the extension $\overline{\mu}$ of μ to S^* is also σ-finite.

12. (a) If f is measurable, then prove that $[x : f(x) = \alpha]$ is measurable for each extended real number α.

Or

(b) Let E be a measurable set. Prove that for each y the set $E + y = [x + y : x \in E]$ is measurable and the measures are the same.

13. (a) Let f and g be integrable functions. Prove the following :

(i) $f + g$ is integrable and

$$\int (f + g)dx = \int fdx + \int gdx.$$

(ii) If A and B are disjoint measurable sets, then

$$\int_A fdx + \int_B fdx = \int_{A \cup B} fdx.$$

Or

(b) Show that $\int_0^1 \sin x \log x \, dx = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)(2n)!}$.

14. (a) Prove that a countable union of sets positive with respect to a signed measured γ is a positive set.

Or

(b) State and prove the Jordan decomposition theorem.

15. (a) If $E \in S \times J$, then prove that for each $x \in X$ and $y \in Y$, $E_x \in J$ and $E^y \in S$.

Or

(b) Show that if μ and γ are σ–finite measures, then $\mu \times \gamma$ is the only measure giving to each measurable rectangle $A \times B$ the measure $\mu(A)\gamma(B)$.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that the Lebesgue outer measure of an interval equals its length.

17. Let c be any real number and let f and g be real-valued measurable functions defined on the same measurable set E. Prove that $f + g$, $f - g$, fg, $f + c$ and cf also measurable.

18. State and prove Fatou's lemma. Using the lemma prove Lebesgue monotone convergence theorem.

19. State and prove the Radon-Nikodym theorem.

20. (a) What is meant by product measure?

(b) State and prove Fubini's theorem.

RW–10627

Sub. Code
511203

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

COMPLEX ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define the analytic function with an example.

2. Define the circle of convergence.

3. Define the rectifiable arc.

4. Define the index of the point.

5. Define the simply connected region with an example.

6. Define an isolated singular point.

7. Define the Harmonic function with an example.

8. Write the Poisson's formula.

9. Define the entire function with an example.

10. Find the value of $\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)$.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) State and prove the Luca's theorem.

Or

(b) State and prove the Abel's limit theorem.

12. (a) Compute $\int_r x\,dz$, where r is the directed line segment from 0 to $1+i$

Or

(b) Prove that a non-constant analytic function maps open sets onto open sets.

13. (a) Prove that a region Ω is simply connected if and only if $n(r, a) = 0$ for all cycles r in Ω and all points a which do not belong to Ω.

Or

(b) State and prove the Rouche's theorem.

14. (a) State and prove the maximum principle for harmonic function.

Or

(b) State and prove the Hurwitz theorem.

15. (a) Express $\sum_{-\infty}^{\infty} \frac{1}{z^3 - n^3}$ in closed form.

Or

(b) Prove that every function which is meromorphic in the whole plane is the quotient of two entire functions.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. State and prove the Abel theorem for radius of convergence of power series.

17. Derive the Cauchy's integral formula.

18. State and prove the Residue theorem.

19. State and prove the Weierstrass theorem.

20. State and prove the Mittag-Leffler theorem.

RW–10628

Sub. Code
511502

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

***Elective* II — NUMERICAL METHODS**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Let u be the tangent direction to a level line $F(x_1, x_2) = \text{constant}$ at a point $x = (x_1, x_2)^T$, show that $(\nabla F)(x)^T u = 0$

2. Write an algorithm for Gauss-Seidal iteration.

3. Distinguish between Chebyshev points and expanded Chebyshev points.

4. Show that, if $p(x)$ is a polynomial of degree less than k, then $p(x)$ is orthogonal to $p_k(x)$, for $k = 2$ only.

5. Write the error term for a formula of $f''(a)$.

6. Evaluate $\int_0^6 \frac{dx}{1+x}$ using Trapezoidal rule taking $h = 1$.

7. Find the general solution of the equation $y' = -2y$.

8. Find an upper bound on the error at $x = 1$ in terms of step size n for the equation $y' = y$, $0 \le x \le 1$, $y(0) = 1$.

9. Derive formula for Adams-Bashforth method.

10. What is the advantages of shooting method?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Use damped Newton's method to find the solution of system $f(\varphi) = 0$ of equations

$$f_1(x) = x_1 + 3 \ln |x_1| - x_2^2, \; f_2(x) = 2x_1^2 - x_1 x_2 - 5x_1 + 1$$

starting with $x^{(0)} = (2, 1)^T$.

Or

(b) Solve for x and y using Newton's method $x^2 + y - 11 = 0$ and $y^2 + x - 7 = 0$, starting with initial approximations $x_0 = 3.5$ and $y_0 = -1.8$.

12. (a) If $F(x) = F(x, c_1, c_2) = c_1 e^{c_2 x}$, derive the normal equations for the best c_1^*, c_2^*. Are these normal equations linear? Justify.

Or

(b) Let the scalar product $\langle g, h \rangle = \int_a^b g(x) h(x) w(x) \, dx$.

Then prove that $P_k(x)$ has k simple real zeroes, all of which lie in the interval (a, b).

13. (a) Derive Simpson's rule in numerical differentiation.

Or

(b) Evaluate $\int_0^5 \frac{1}{4x+5}\,dx$ by three-point Gaussian formula.

14. (a) Find the general solution of the difference equation $y_{n+2} - 4y_{n+1} + 3y_n = 2^n$, $y_0 = 0$, $y_1 = 1$.

Or

(b) Using Taylor's series, find the solution of the differential equation $xy' = x - y$, $y(2) = 2$ at $x = 2.1$ correct to five decimal places.

15. (a) Solve $\frac{dy}{dx} = x - \frac{1}{y}$, $y(0) = 1$ from $x = 0$ to $x = 0.2$, using second-order predictor-corrector method with $h = 0.1$.

Or

(b) Use finite difference method to solve for $y(x)$ given $\frac{d^2y}{dx^2} + y + 1 = 0$, $x \in (0, 1)$, $y(0) = 0$, $y(1) = 0$, taking $h = \frac{1}{4}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Solve the non-linear system $x^2 + xy^3 = 9$, $3x^2y - y^3 = 4$ using the fixed point iteration method.

17. If $f(x) = 6000 + x$, find the least-squares approximation to $f(x)$ by straight lines. Also find the polynomial $p_1^*(x) = d_0^* + d_1^* x$ which minimizes $\sum_{n=-2}^{2} [f(n) - p_1(n)]^2$.

18. Construct a rule of the form $I(f) = \int_{-1}^{1} f(x)\, dx \equiv A_0 f\left(-\frac{1}{2}\right) + A, \quad f(0) + A_2 f\left(\frac{1}{2}\right)$ which is exact for all polynomials of degree S^2.

19. Use Runge-Kutta method of fourth order to find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$, taking $h = 0.1$.

20. Determine the value of $y(0.4)$ by Milne's method given $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$, use Taylor series to get the values of $y(0.1)$, $y(0.2)$ and $y(0.3)$.

RW–10629

Sub. Code
511401

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

PROBABILITY AND STATISTICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Let the random variable X have the p.d.f. $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$

   Find $\Pr\left(-\frac{1}{2} < X < \frac{1}{2}\right)$.

2. Let X and Y have the joint p.d.f

   $f(x, y) = \begin{cases} x + y, & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$

   Find $E(XY^2)$.

3. Let X and Y have the joint p.d.f

   $f(x, y) = \begin{cases} 2, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$

   Find the marginal density function.

4. Define correlation coefficient.

5. If X is $b(n, p)$, show that $E\left(\frac{X}{n}\right) = P$.

6. If X is $n(\mu, \sigma^2)$ find 'b' so that $\Pr\left(-b < \frac{(x-\mu)}{\sigma} < b\right) =$ 0.90.

7. Define mean and variance of random variable.

8. Determine the constant K such that, $f(x) = \begin{cases} Kx\ (3-4)^4 & \text{for } 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$ is a p.d.f.

9. Let X_1 and X_2 have a trinomial distribution with parameter n, P_1, P_2. Find the distribution of $Y = X_1 + X_2$.

10. Let $\overline{X}_n$ denote the mean of a random sample of size n from a distribution that is $n(\mu, \sigma^2)$. Find the limiting distribution of $\overline{X}_n$.

Part B (5 × 5 = 25)

Answer **all** questions

11. (a) Let C_1 and C_2 be subsets of $\mathcal{C}$. Prove that

(i) if $C_1 \subset C_2$, then $P(C_1) \le P(C_2)$

(ii) $P(C_1 \cup C_2) = P(C_1) + P(C_2) - p(C_1 \cap C_2)$.

Or

(b) State and prove Chebyshev's inequality.

12. (a) Prove that the correlation co-efficient is independent of change of origin and scale.

Or

(b) Let the joint p.d.f of X and Y be

$$f(x, y) = \begin{cases} 12\,xy\,(1-y) & 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Examine whether X and Y are stochastically independent.

13. (a) Compute the measures of skewness and kurtosis of the binomial distribution

Or

(b) Prove that, if the random variable X is $n(\mu, \sigma^2)$, $\sigma^2 > 0$, then the random variable $V = \dfrac{(X-\mu)^2}{\sigma^2}$ is $\chi^2(1)$.

14. (a) Derive the p.d.f. of t–distribution and obtain its mean.

Or

(b) Let $X_1, \ldots\ldots X_n$ be mutually stochastically independent random variables having, respectively the normal distributions $n(\mu_1, \sigma_1^2)$, $n(\mu_2, \sigma_2^2), \ldots.$ $n(\mu_n, \sigma_n^2)$. Prove that the random variable

$Y = K_1X_1 + ... + K_nX_n$, where $K_1,....K_n$ are real constants, is normally distributed with mean $K_1\,\mu_1 + ... + K_n\mu_n$ and variance $K_1^2\ \sigma_1^2 + ... + K_n^2\,\sigma_n^2$.

(i.e.) Y is $n\left(\sum_1^n K_i\mu_i,\ \sum_1^n K_i^2\,\sigma_i^2\right)$.

15. (a) Let X be $n(\mu, \sigma^2)$ and consider the transformation $X = \ln Y$ or $Y = e^x$ (equivalently),

(i) Find the p.d.f. of Y

(ii) Find the mean and the variance of Y by first determining $E(e^X)$ and $E[(e^X)^2]$.

Or

(b) Let $\overline{X}_n$ denote the mean of a random sample of size n from a Poisson distribution with parameter $\mu = 1$. Show that the m.g.f. of $Y_n = \sqrt{n}\,\dfrac{(\overline{X}_n - \mu)}{\sigma} = \sqrt{n}\,(\overline{X}_n - 1)$ is given by exp $[-t\sqrt{n} + n(e^{t\sqrt{n}} - 1)]$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) State and prove Baye's theorem.

(b) A random variable X has a p.d.f.

$$f(x)=\begin{cases}\dfrac{3!}{x!(3-x)!}\left(\dfrac{1}{2}\right)^3, & x = 0, 1, 2, 3\\ 0 & \text{otherwise}\end{cases}$$

Find the mean and variance.

17. Let $f\left(\dfrac{x_1}{x_2}\right)=\dfrac{C_1x_1}{x_2^2}$, $0 < x_1 < x_2$, $0 < x_2 < 1$. Zero elsewhere, and $f_2(x_2)=C_2x_2^4$, $0 < x_2 < 1$, zero elsewhere denote, respectively, the conditional p.d.f of X_1, given $X_2 = x_2$, and marginal p.d.f of X_2 .

Determine :

(a) the constants C_1 and C_2

(b) the joint p.d.f. of X_1 and X_2

(c) $\Pr\left(\dfrac{1}{4} < X_1 < \dfrac{1}{2} / X_2 = \dfrac{5}{8}\right)$

(d) $\Pr\left(\dfrac{1}{4} < X_1 < \dfrac{1}{2}\right)$.

18. (a) State and prove the recurrence relation for the Poisson distribution.

(b) If X is $n(\mu, \sigma^2)$, show that $E(X-\mu)=\sigma\sqrt{2}/\pi$.

19. (a) S.T $S^2 = \frac{1}{n}\sum_{1}^{n}(X_i - \overline{X})^2$ where $\overline{X} = \sum_{1}^{n}\frac{X_i}{n}$.

(b) Let X_1 and X_2 be stochastically independent with normal distributions $n(\mu_1, \sigma^2)$ and $n(\mu_2, \sigma_2^2)$ respectively. Define the random variable $Y = X_1 - X_2$. Find $g(y)$ the p.d.f of Y.

20. State and prove central limit theorem.

RW–10630

Sub. Code
511402

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

FUNCTIONAL ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define a normed space. Give an example.

2. Define : Bounded linear operator and inverse operator.

3. Write down the polarization identity.

4. Prove that an orthonormal set is linearly independent.

5. What is meant by serquilinear functional?

6. With usual notations, prove that $O^* = 0, I^* = I$.

7. Show that a norm on a vector space X is a sublinear functional on X.

8. Define the following terms :

   (a) rare

   (b) meager

   (c) nonmeager.

9. What do you mean by weak* convergence of a sequence of functionals?

10. State open mapping theorem.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Let $\{x_1, x_2, ..., x_n\}$ be a linearly independent set of vectors in a normed space *X*. Prove that for every choice of scalars $\alpha_1, \alpha_2, ..., \alpha_n$, there is a number $c > 0$ such that
$\|\alpha_1 x_1 + \alpha_2 x_2 + ... + \alpha_n x_n\| \geq e(|\alpha_1| + |\alpha_2| + ... + |\alpha_n|)$.

Or

(b) With usual notations, prove that

(i) $\|T_1 T_2\| \leq \|T_1\| \|T_2\|$;

(ii) $\|T^n\| \leq \|T\|^n$.

12. (a) State and prove Schwarz inequality.

Or

(b) Let (e_k) be an orthonormal sequence in an inner product space *X*. Prove that $\sum_{K=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$.

13. (a) Show that the dual space of the real space l^2 is l^2.

Or

(b) Let the operators $U : H \to H$ and $V : H \to H$ be unitary (Here, *H* is a Hilbert space). Prove that

(i) $\|U\| = 1$, provided $H \neq \{0\}$;

(ii) *U* is normal.

14. (a) State and prove the norm of the adjoint operator theorem.

Or

(b) Prove that the normed space X of all polynomials with norm defined by $\|x\| = \max_j |\alpha_j|, (\alpha_0, \alpha_1, \ldots$ the coefficients of x) is not complete.

15. (a) Let X and Y be normed spaces, $T \in B(X,Y)$ and (x_n) a sequence in X. If $x_n \xrightarrow{w} x_0$, then prove that $Tx_n \xrightarrow{w} Tx_0$.

Or

(b) Let $T_n \in B(X,Y)$, where X is a Banach space and Y a normed space. If (T_n) is strongly operator convergent with limit T, then prove that $T \in B(X,Y)$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Prove that a finite dimensional vector space is algebraically reflexive.

(b) Show that the dual space of l^1 is l^∞.

17. State and prove separable Hilbert spaces theorem.

18. State and prove the Riesz's theorem for the functionals on Hilbert spaces.

19. State and prove generalized Hahn-Banach theorem.

20. (a) State and prove closed graph theorem.

    (b) If the inverse T^{-1} of a closed linear operator exists, then prove that T^{-1} is a closed linear operator.

RW–10631

Sub. Code
511402

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Mathematics

COMPLEX ANALYSIS

(2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find the smallest positive period of e^z.

2. Define indirectly conformal mapping.

3. Define rectifiable arcs.

4. Compute $\int_{|z|=2} \frac{dz}{z^2+1}$.

5. Explain $r \sim 0(\text{mod}\,\Omega)$ where r is a cycle.

6. Find the pole and residue of $\cot z$.

7. Write the imaginary part of u which has a conjugate harmonic function v.

8. Explain the Laurent series represents an analytic function in $|z| > R$.

9. Write the necessary condition for convergence for infinite products.

10. Show that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) State and prove Luca's theorem.

Or

(b) Illustrate that the mapping of the whole region Ω is necessarily topological.

12. (a) Explain the complex line integral of $f(z)$ over the arc r.

Or

(b) Necessary and sufficient condition under which a line integral depends only on the end points.

13. (a) A region Ω is simply, connected if and only if $n(r, a) = 0$ for all cycles γ in Ω and all points $a \neq \Omega$

Or

(b) If $f(z)$ is monomorphic in Ω with the zeros a_j, and the poles b_k, prove that

$$\frac{1}{2\pi i}\int_{\gamma}\frac{f'(z)}{f(z)} = \sum_j n(\gamma, a_j) - \sum_k n(\gamma, b_k).$$

14. (a) Derive Poisson's formula.

Or

(b) If $f(z)$ is analytic in the region Ω containing z_0, then prove that

$$f(z) = f(z_0) + \frac{f'(z_0)}{1!}(z - z_0) + \cdots + \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n + \cdots$$

is valid in the largest open disk of center z_0 in Ω.

15. (a) State and prove Mittag-Leffler theorem of generalization to arbitrary regions.

Or

(b) A necessary and sufficient condition for the absolute convergence of $\prod_{1}^{\infty}(1+a_n)$ is the convergence of the series $\sum_{1}^{\infty}|a_n|$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that a linear transformation carries circles into circles.

17. State and prove Cauchy's theorem.

18. If $pdx + qdy$ is locally exact in Ω then prove that $\int_{\gamma} pdx + qdy = 0$ for every cycle $\gamma \sim 0$ in Ω.

19. State and prove Weierstrass's theorem.

20. Prove that every function which is monomorphic in the whole plane is the quotient of two entire functions.

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RW–10672

Sub. Code
511101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

ALGEBRA-I

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 ×2 = 20)

Answer **all** questions.

1. How many generators does a cyclic group of order n have?

2. Find the number of group homomorphism from Z_{12} to Z_{18}.

3. Find the number of conjugate class of S_6.

4. Show that any group of order 15 is cyclic.

5. Write the units in Z_{11}.

6. Find the number of ring homomorphism from Z_{10} to Z_{22}.

7. Is the polynomial x^2+1 is irreducible over Z_{11}.

8. Given an example of Euclidean ring which is not a field.

9. Find the number of non-isomorphic abelian groups of order 25.

10. Define modules with example.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) If H and K are subgroup of G, prove that HK is a subgroup of G if and only if $HK=KH$.

Or

(b) If φ is a homomorphism of G into G' with kernel K, then K is a normal subgroup of G.

12. (a) If $o(G) = p^n$ where p is a prime number, prove that $Z(G) \neq (e)$.

Or

(b) If p is a prime number and $p \mid o(G)$, Show that G has an element of order p.

13. (a) Prove that a finite integral domain is a field.

Or

(b) Let R be a commutative ring with unit element whose only ideals are (0) and R Itself, prove that R is a field.

14. (a) State and prove unique factorization theorem.

Or

(b) Let J[i] denote the set of all complex numbers of the form $a+bi$ where a and b are integers. Prove that J[i] is a Euclidean ring.

15. (a) Let R be a Euclidean ring, prove that any finitely-generated R-module, M, is the direct sum of a finite number of cyclic submodules.

Or

(b) If A and B are submodules of M prove that $A \cap B$ is a submodule of M.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that every group is isomorphic to a subgroup of $A(S)$ for some appropriate S.

17. State and prove Sylow's third theorem.

18. Prove that every integral domain can be imbedded in a field.

19. State and prove the Eisenstein Criterion.

20. Let $p(n)$ be the number of partitions of n. Prove that number of non-isomorphic abelian groups of order p^n is $p(n)$.

RW–10673

Sub. Code
511102

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

REAL ANALYSIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define perfect set.

2. Define set.

3. Define continuous at a point.

4. Define monotonically increasing function.

5. Define differentiable function.

6. What is meant by local maximum?

7. Define partition.

8. State fundamental theorem of calculus.

9. Define uniformly convergence.

10. Define uniformly bounded.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) If $x \in R$, $y \in R$ and $x > 0$, prove that there is a positive integer n such that $nx > y$.

Or

(b) Prove that compact subsets of metric spaces are compact.

12. (a) Prove that a mapping f of a metric space X into a metric space Y is continuous on X iff $f^{-1}(V)$ is open in X for every open set V in Y.

Or

(b) Suppose f is a continuous mapping of a compact metric space X into a metric space Y. Prove that $f(x)$ is compact.

13. (a) State and prove Taylor's theorem.

Or

(b) State and prove mean value theorem.

14. (a) Prove that $f \in \Re(\alpha)$ on [a, b] iff for every $\varepsilon > 0$ there exists a partition P such that $U(P, f, \alpha) - L(p, f, \alpha) < \varepsilon$.

Or

(b) Suppose $f \in \Re(\alpha)$ on $[a, b]$, $m \le f \le M, \phi$ is continuous on [m, M] and $h(x) = \phi f(x))$ on $[a, b]$. Prove that $h \in \Re(\alpha)$ on $[a, b]$.

15. (a) Prove that the sequence of functions $\{f_n\}$, Define on E, converges uniformly on E iff for every $\varepsilon > 0$ there exists an integer N such that $m \geq N$, $n \geq N$, $x \in E$ implies $|f_n(x) - f_m(x) \leq \varepsilon|$.

Or

(b) Let α be monotonically increasing on $[a,b]$. Suppose $f_n \in \Re(\alpha)$ on $[a,b]$, for $n = 1,2,3....$ and suppose $f_n \to f$ uniformly on $[a,b]$, prove that $f \in \Re(\alpha)$ on $[a,b]$ and $\int_a^b f d\alpha = \lim_{n\to\infty} \int_a^b f_n d\alpha$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that for every real $x > 0$ and every integer $n > 0$ there is one and only one real y such that $y^n = x$.

17. If Let f be a continuous mapping of a metric space X into a metric space Y and if E is a connected subset of X. Prove that $f(E)$ is connected.

18. State and prove L'Hospital's rule.

19. Assume α increases monotonically and $\alpha^1 \in \Re$ on $[a,b]$. Let f be a bounded real function on $[a,b]$. Prove that $f \in \Re(\alpha)$ iff $f\alpha^1 \in \Re$. In that case $\int_a^b f d\alpha = \int_a^b f(x)\alpha^1(x)\,dx$.

20. State and prove Stone-Weierstrass theorem.

RW–10674

Sub. Code
511103

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

DIFFERENTIAL EQUATIONS

(2011 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Each question carries 2 marks.

1. Write the statement of Existence theorem.

2. Find two linearly independent power series solutions (in powers of x) of the equation $y'' - xy' + y = 0$.

3. Write the second order Euler equation.

4. Find the solutions of the following equation for $x > 0$: $x^2y'' + 2xy' - 6y = 0$.

5. Find the integral surface of the linear partial differential equation $x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$ which contains the straight line $x + y = 0$, $z = 1$.

6. Find a complete integral of the equation $p^2x + q^2y = z$.

7. Write the one-dimensional wave equation.

8. If $\alpha_r D + \beta_r D' + \gamma_r$ is a factor of $F(D, D')$ and $\phi_r(\xi)$ is an arbitrary function of the single variable ξ prove that if $\alpha_r \neq 0$ $u_r = \exp\left(\frac{\gamma_r x}{\alpha_r}\right) \phi_r(\beta_r x - \alpha_r y)$.

9. Write the Helmholz's equation.

10. Write the d'Alembert's solution.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

Each question carries 5 marks.

11. (a) State and prove value range theorem.

Or

(b) Let $\alpha_1, \ldots, \alpha_n$ any n constants and let x_0 be any real number. On any interval I containing x_0 prove that there exits at most one solution ϕ of $L(y) = 0$ satisfying $\phi(x_0) = \alpha_1$, $\phi'(x_0) = \alpha_2, \ldots, \phi^{(n-1)}(x_0) = \alpha_n$.

12. (a) Find all solutions of the following equations for $|x| > 0$.

(i) $x^2 y'' + xy' + 4y = 1$

(ii) $x^2 y'' - 3xy' + 5y = 0$.

Or

(b) Compute the indicial polynomials and their roots, for the following equations :

(i) $x^2 y'' + (x + x^2) y' - y = 0$

(ii) $4x^2 y'' + (\sin x) y' + (\cos x) y = 0$.

13. (a) Find the general solution of the differential equation $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = (x+y)z$.

Or

(b) Find a complete integral of the equation $p^2 x + q^2 y = z$.

14. (a) If u is the complementary function and z_1 a particular integral of a linear partial differential equation, show that $u + z_1$ is a general solution of the equation.

Or

(b) Reduce the equation $\frac{\partial^2 z}{\partial x^2} = x^2 \frac{\partial^2 z}{\partial y^2}$ to canonical form.

15. (a) If $\rho > 0$ and $\psi(r) = \int_S \frac{\sigma(r')dS'}{|r - r'|}$, where the volume V is bounded, prove that $\lim_{r=\infty} r\psi(r) = M$ where $M = \int_V \rho(r')dr'$.

Or

(b) If the string is released from rest in the position $y = \frac{4\in}{l^2} x(1-x)$ show that its motion is described by the equation $y = \frac{32\in}{3} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} \sin\frac{(2n+1)\pi x}{l} \cos\frac{(2n+1)\pi ct}{l}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. Prove that, if $\phi_1, \phi_2, .., \phi_n$ are n solutions of $L(y) = 0$ on an interval I, they are linearly independent there if and only if $W(\phi_1, \phi_2, .., \phi_n)(x) \neq 0$ for all x in I.

17. Find the solution ϕ of the form $\phi(x) = x^r \sum_{k=0}^{\infty} c_k x^k$, $(x > 0)$ for the following equations :

    (a) $2x^2 y'' + (x^2 - x) y' + y = 0$.

    (b) $x^2 y'' + (x - x^2) y' + y = 0$.

18. (a) Find the integral surface of the equation $(x - y) y^2 p + (y - x) x^2 q = (x^2 + y^2) z$ through the curve $xz = a^3$, $y = 0$.

    (b) Find the general integral of the equation $(x - y) p + (y - x - z) q = (z + a) z$ are the particular solution through the circle $z = 1$, $x^2 + y^2 = 1$.

19. Reduce the following equations to canonical form and find its general solution :

    (a) $(n-1)^2 \frac{\partial^2 z}{\partial x^2} - y^{2n} \frac{\partial^2 z}{\partial y^2} = n y^{2n-1} \frac{\partial z}{\partial y}$.

    (b) $\frac{\partial^2 z}{\partial x^2} + 8 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$.

20. Explain the elementary solutions of the one-dimensional wave equation.

RW–10675

Sub. Code
511501

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Mathematics

ANALYTICAL NUMBER THEORY

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. If $(a,b) = (a,c) = 1$, then prove that $(a,bc) = 1$.

2. State true or false, Justify your assertion. If the same primes divide m and n, then $n\phi(m) = m\phi(n)$.

3. What is the average order of $\mu(n)$ and $\wedge(n)$?

4. Define two mutually visible lattice points.

5. Given two functions $f(x)$ and $g(x)$, when do you say that $f(x) = 0(g(x))$.

6. Give a lower bound for $\pi(n)$ that holds for every integer $n \geq 2$.

7. Prove that congruence is an equivalence relation.

8. Give a Fermat number which is also prime.

9. Find the quadratic residues modulo 11.

10. Define Legendre's symbol (n/p). Find the value of $(22/11)$.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

Each question carries 5 marks.

11. (a) Prove that there are infinitely many prime numbers.

Or

(b) Prove that for $n \geq 1$, we have $\phi(n) = n\prod_{p/n}\left(1 - \frac{1}{P}\right)$

12. (a) State and prove Euler's summation formula.

Or

(b) Prove that the set of lattice points visible from the origin has density $6/\pi^2$.

13. (a) Derive Abel's identity.

Or

(b) For $n \geq 1$, prove that the nth prime P_n satisfies the inequalities $\frac{1}{6}\ n\log n < P_n < 12\left(n\log n + n\log\frac{12}{e}\right)$.

14. (a) Assume $(a,m)=d$ and suppose that d/b . Prove that the linear congruence $ax \equiv b \pmod{m}$ has exactly d solutions modulo m.

Or

(b) State and prove Lagrange's theorem.

15. (a) Prove that for every odd prime P, we have $(2/P)=(-1)^{(P^2-1)/8}=\begin{cases}1 & \text{if } p \equiv \pm 1 \pmod 8 \\ -1 & \text{if } p \equiv \pm 3 \pmod 8\end{cases}$.

Or

(b) If P is an odd positive integer, prove that $(-1/P)=(-1)^{(P-1)/2}$ and $(2/P)=(-1)^{(P^2-1)/8}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

Each question carries 10 marks.

16. State and prove the division algorithm. Hence state and prove the euclidean algorithm.

17. For $x \geq 2$, prove that $\sum_{p \leq x}\left[\frac{x}{P}\right]\log p = x \log x + o(x)$, where the sum is extended over all primes $\leq x$.

18. Prove that the following relations are logically equivalent

(a) $\lim_{x\to\infty} \frac{\pi(x)\log x}{x} = 1$ (b) $\lim_{x\to\infty} \frac{\theta x}{x} = 1$ (c) $\lim_{x\to\infty} \frac{\psi x}{x} = 1$.

19. State and prove the principle of cross – classification.

20. State and prove the Gauss' lemma.

RW–10454

Sub. Code
511101

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Mathematics

ALGEBRA — I

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

Each question carries 2 marks.

1. Let H be a subgroup of a finite group G. Prove that $|Ha| = |H| \forall a \in G$.

2. If H and K are subgroups of G and $O(H) > \sqrt{O(G)}$, $O(K) > \sqrt{O(G)}$, prove that $H \cap \neq \{e\}$.

3. Show that the center of a group is a normal subgroup.

4. Let ϕ be an automorphism of G. If $a \in G$ and $o(a) > 0$, prove that $o(\phi(a)) = o(a)$.

5. Show that any group of order 15 is cyclic.

6. How many abelian groups of order 8 exists?

7. Find all units in $(Z, +_8, \cdot_8)$

8. Find a maximal ideal in $(Z, +, \cdot)$

9. Define Euclidean ring. Give an example.

10. State the Eisenstein criterion.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Show that, for $a, b \in G$, $ax = b$ and $ya = b$ are uniquely solvable in G.

Or

(b) Suppose H is the only subgroup of order $o(H)$ in the finite group G, prove that H is a normal subgroup of G.

12. (a) State and prove Fundamental Homomorphism theorem.

Or

(b) Show that A_n is a normal subgroup of S_n.

13. (a) Prove that every group of order p^2, where p is a prime number is abelian.

Or

(b) Derive class equation for a finite group.

14. (a) Prove that the ring of integer mod p is a field if and only if p is prime number.

Or

(b) Show that the homomorphism ϕ of R into R' is an isomorphism if and only if $\ker \phi = \{o\}$.

15. (a) Let R be a Euclidean ring. Prove that every element in R is either a unit in R or can be written as the product of a finite number of prime elements of R.

Or

(b) If R is an integral domain, prove that $R[x]$ is an integral domain.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. If H and K are finite subgroups of G, prove that $O(HK) = \dfrac{O(H)\,O(K)}{O(H \cap K)}$.

17. State and prove Cauchy's theorem for abelian groups.

18. State and prove the first past of Sylow's theorem.

19. Prove that every integral domain can be embedded in a field.

20. Let R be a commutative ring with unit element. Prove that an ideal M of R is maximal if and only if $\dfrac{R}{M}$ is a field.